

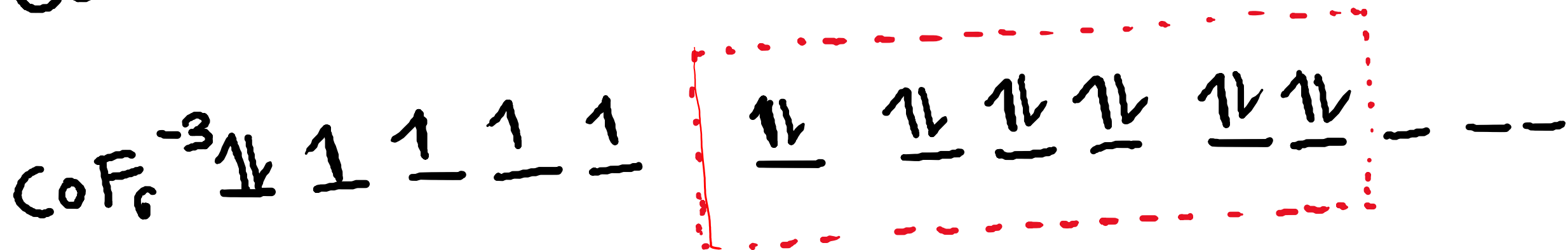
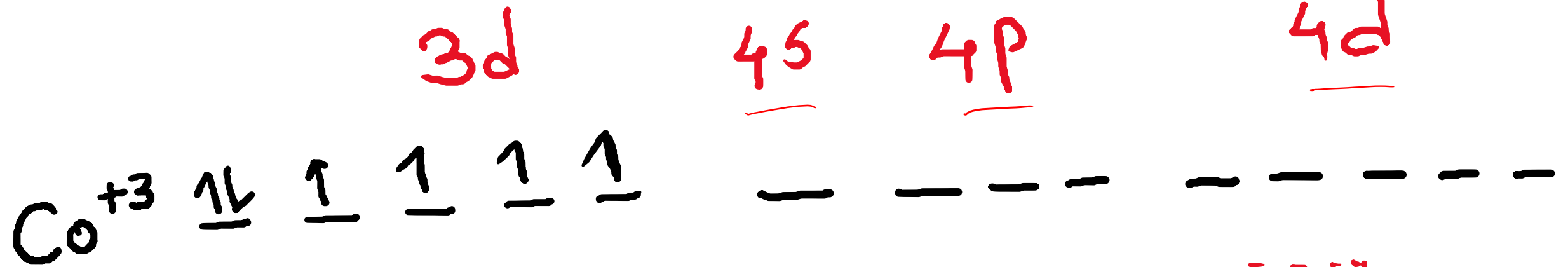
Bonding in Coordination Compounds

Valence Bond Theory

This theory was developed by Pauling. The model utilizes hybridization of metal valence orbitals to account for the observed structures and magnetic properties of complexes.

Pauling suggested that $(n-1)d$, ns and np undergo hybridization to give hybridized orbitals.

An empty hybrid orbital on the metal center can accept a pair of electrons from a ligand to form a σ -bond.





5d

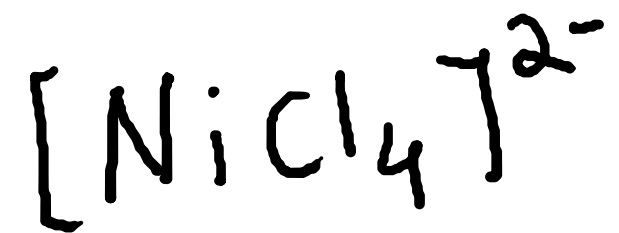


6s



6p

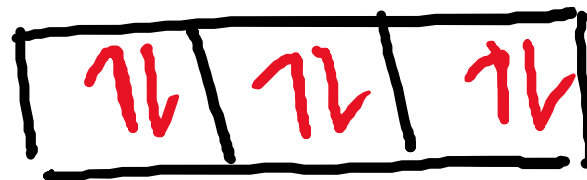
dsp^2



3d



4s



4p

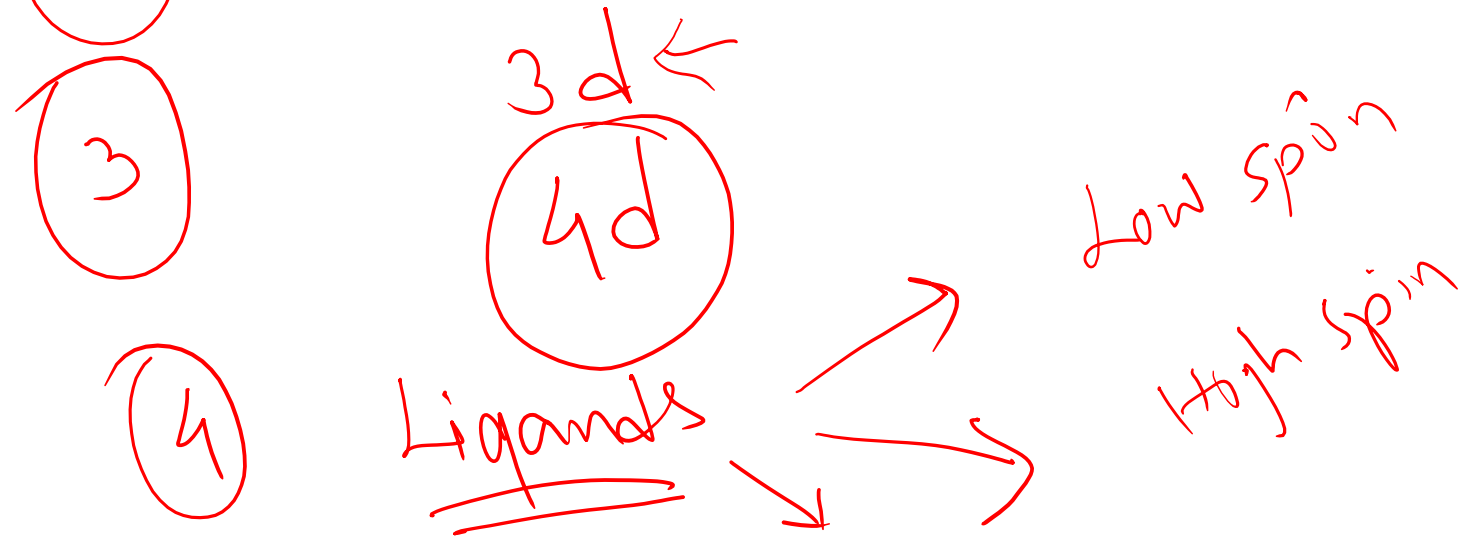


C.N.	Geometry	Description	Examples
2	Linear	sp	$[\text{Ag}(\text{NH}_3)_2]^+$
3	Trigonal planar	sp^2	$[\text{HgI}_3]^-$
4	Tetrahedral	sp^3	$[\text{FeBr}_4]^{2-}$
5	Square planar	sp^2d	$[\text{Ni}(\text{CN})_4]^{2-}$
4	Trigonal bipyramidal	sp^3d	$[\text{CuCl}_5]^{3-}$
5	Square-based pyramidal	sp^3d	$[\text{Ni}(\text{CN})_5]^{3-}$
6	Octahedral	sp^3d^2	$[\text{Co}(\text{NH}_3)_6]^{3+}$

Limitations

-1

- ① Electronic Spectroscopy
- ② "High spin - Low Spin" complexes



Crystal Field Theory

In view of the weaknesses of VBT, an alternative bonding model was applied to transition metal complexes. This is known as crystal field theory (CFT).

It has been originally proposed by Hans Bethe and Van Vleck in 1929.

The assumptions of the crystal field theory are as follows:

The interaction between the metal ion and the ligand is assumed to be purely electrostatic (ionic) in nature.

The ligands are treated as point charges.

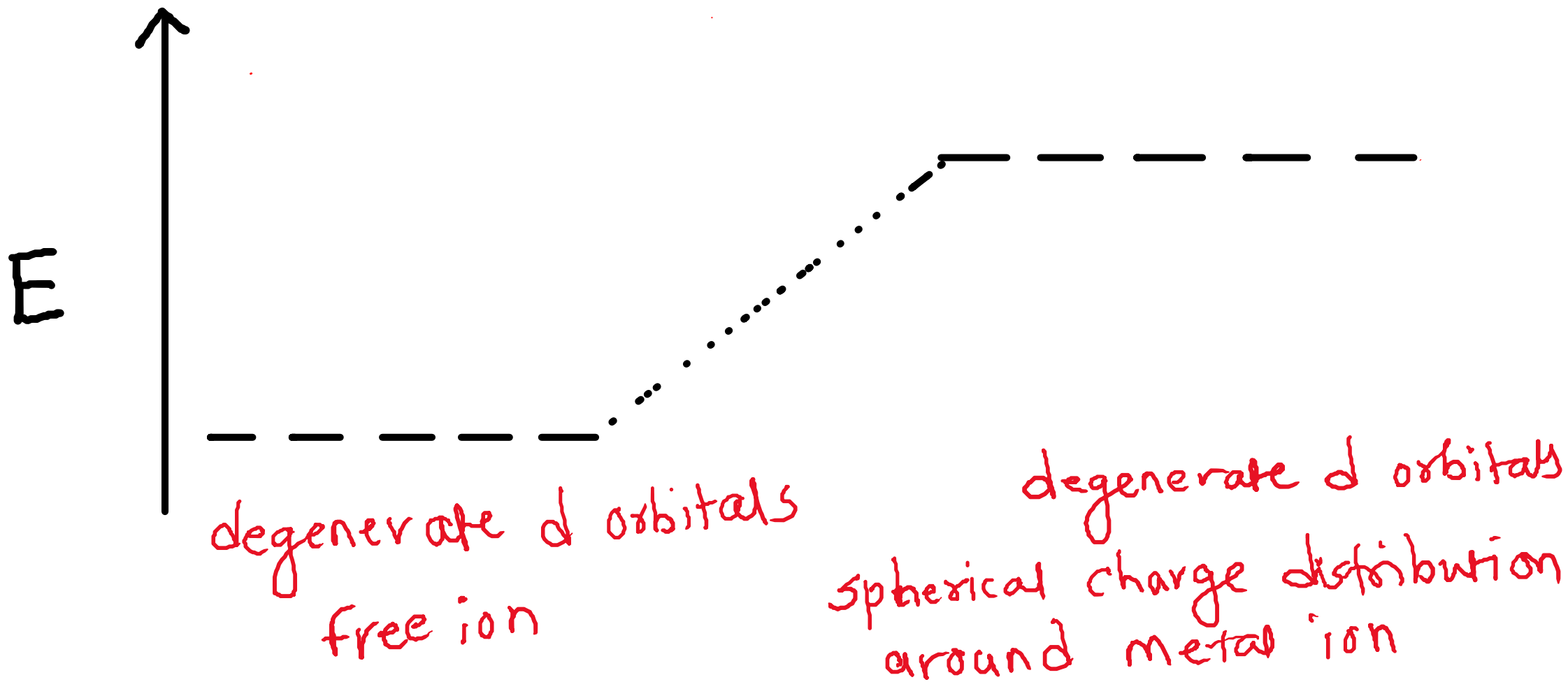
Consequent to this assumption, the CFT does not allow for any overlap between the metal orbitals and the ligand orbitals.

Consequences of the metal ion interaction with ligands

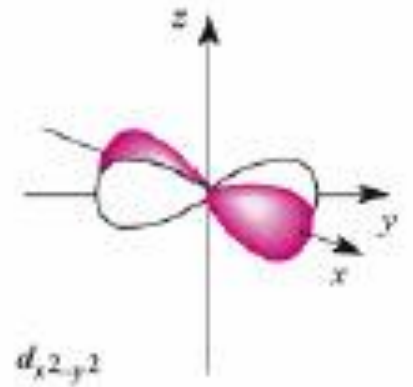
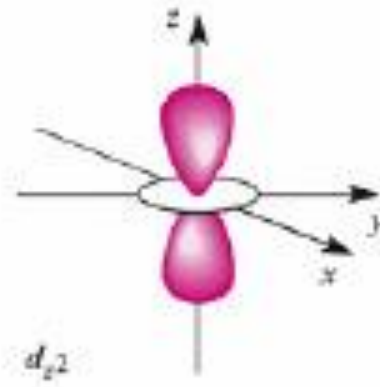
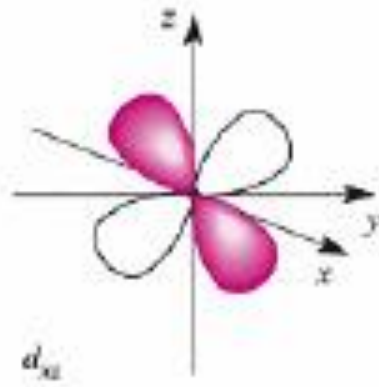
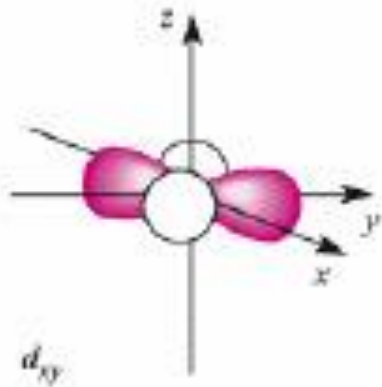
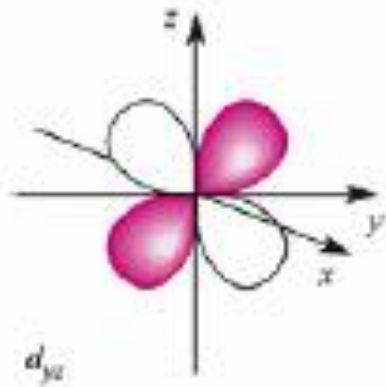
The five d orbitals in an isolated, gaseous metal ion are degenerate (are of the same energy and symmetry).

If a spherically symmetric field of negative charges is placed around the metal ion the orbitals will remain degenerate, but all of them will be raised in energy as a result of repulsion between the negative field and the metal electrons in the orbitals.

If the field results produced by the ligands is not spherically symmetric but of a particular symmetry (octahedral, tetrahedral etc), the d-orbitals are not affected equally and the degeneracy of the d orbitals is removed.



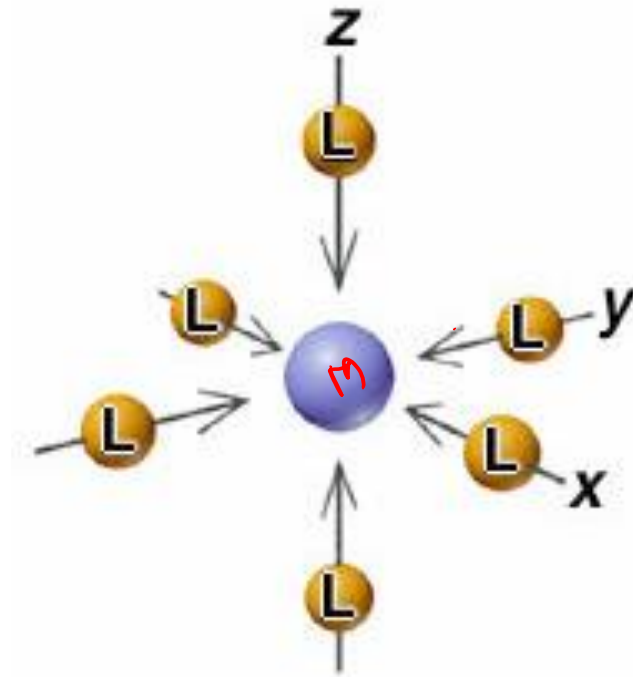
Shape of d orbitals



Octahedral Field

An octahedral field is described by six ligands at the six corners of an octahedron and the metal is at the center.

Hence, the orbitals lying along the axes are strongly repelled than those whose orbitals are lying in between the axes (d_{xz} , d_{yz} , d_{xy}).

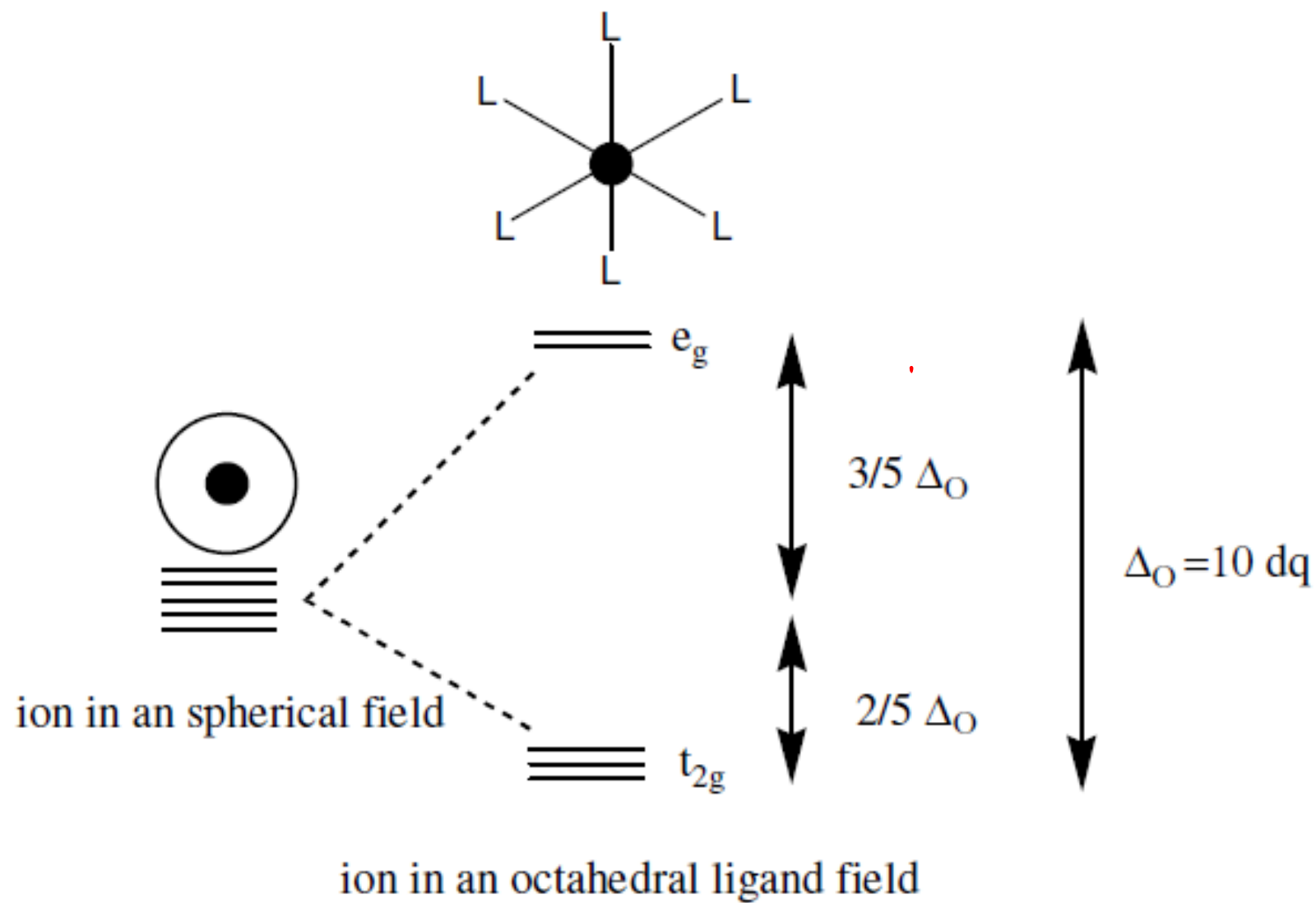


The higher energy set of orbitals (d_{z^2} and $d_{x^2-y^2}$) are labeled as e_g

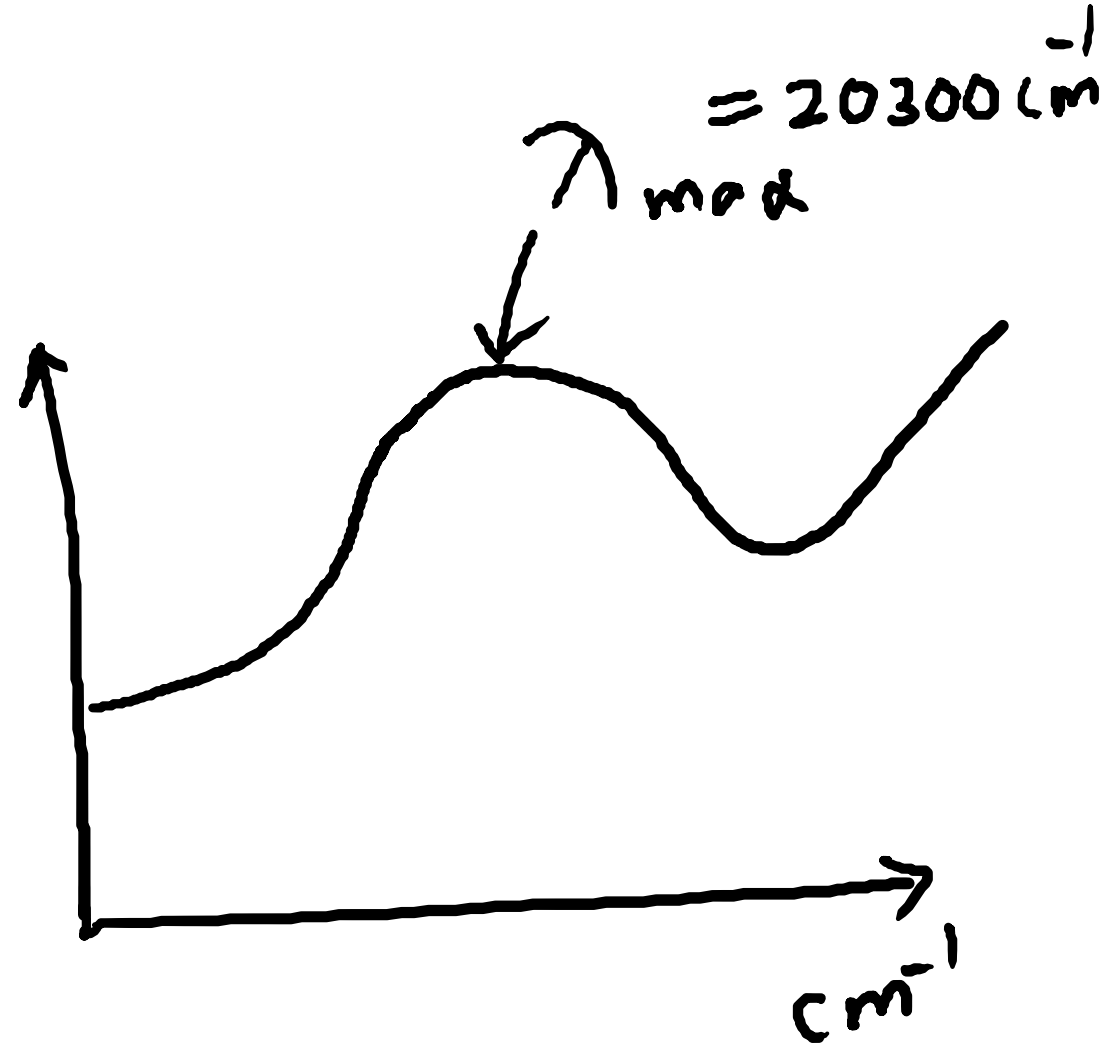
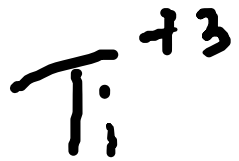
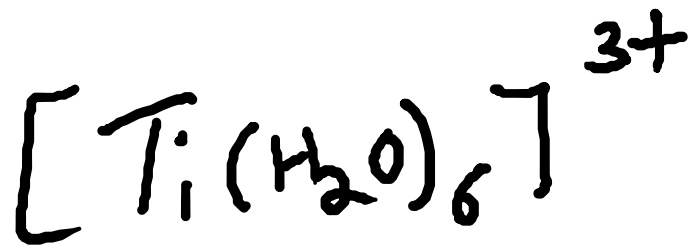
and the lower energy set is labeled as t_{2g}).

The energy separation between the two levels is denoted by Δ_o or $10 Dq$.

Splitting of d-orbitals in an Octahedral field



Splitting Energy Δ_o



Doct → Experimental quantity

Complex	Δ / cm^{-1}	Complex	Δ / cm^{-1}
$[\text{TiF}_6]^{3-}$	17 000	$[\text{Fe}(\text{ox})_3]^{3-}$	14 100
$[\text{Ti}(\text{OH}_2)_6]^{3+}$	20 300	$[\text{Fe}(\text{CN})_6]^{3-}$	35 000
$[\text{V}(\text{OH}_2)_6]^{3+}$	17 850	$[\text{Fe}(\text{CN})_6]^{4-}$	33 800
$[\text{V}(\text{OH}_2)_6]^{2+}$	12 400	$[\text{CoF}_6]^{3-}$	13 100
$[\text{CrF}_6]^{3-}$	15 000	$[\text{Co}(\text{NH}_3)_6]^{3+}$	22 900
$[\text{Cr}(\text{OH}_2)_6]^{3+}$	17 400	$[\text{Co}(\text{NH}_3)_6]^{2+}$	10 200
$[\text{Cr}(\text{OH}_2)_6]^{2+}$	14 100	$[\text{Co}(\text{en})_3]^{3+}$	24 000
$[\text{Cr}(\text{NH}_3)_6]^{3+}$	21 600	$[\text{Co}(\text{OH}_2)_6]^{3+}$	18 200
$[\text{Cr}(\text{CN})_6]^{3-}$	26 600	$[\text{Co}(\text{OH}_2)_6]^{2+}$	9 300
$[\text{MnF}_6]^{2-}$	21 800	$[\text{Ni}(\text{OH}_2)_6]^{2+}$	8 500
$[\text{Fe}(\text{OH}_2)_6]^{3+}$	13 700	$[\text{Ni}(\text{NH}_3)_6]^{2+}$	10 800
$[\text{Fe}(\text{OH}_2)_6]^{2+}$	9 400	$[\text{Ni}(\text{en})_3]^{2+}$	11 500

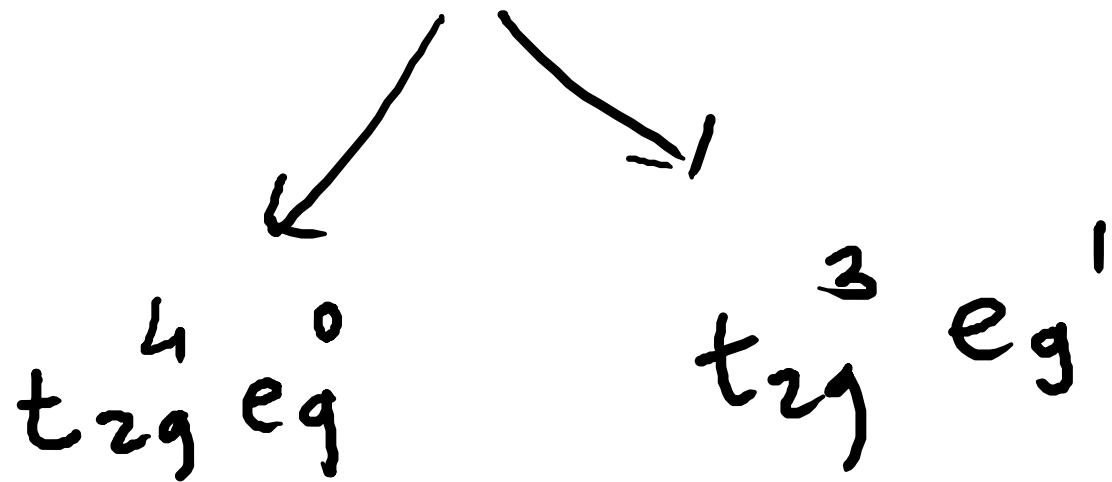
CFSE

$$d^1 \longrightarrow 0.4 \Delta_0$$

$$d^2 \longrightarrow 0.8 \Delta_0$$

$$d^3 \longrightarrow 1.2 \Delta_0$$

for d^4 two possibilities



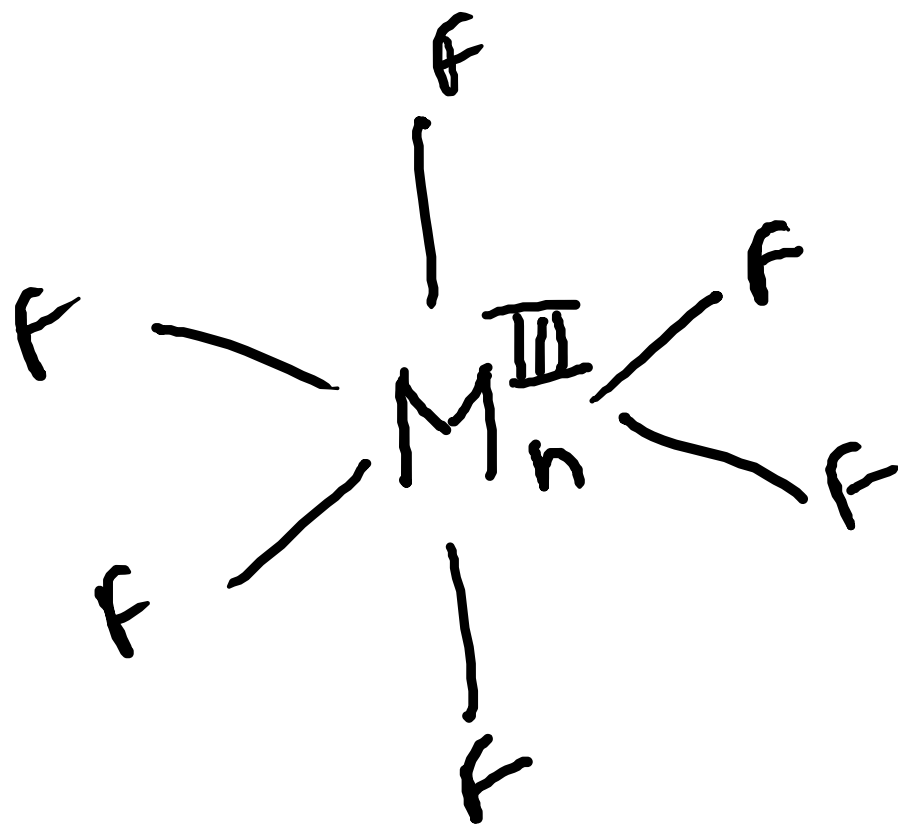
pairing energy $\rightarrow$ Coulombic repulsion
Exchange energy

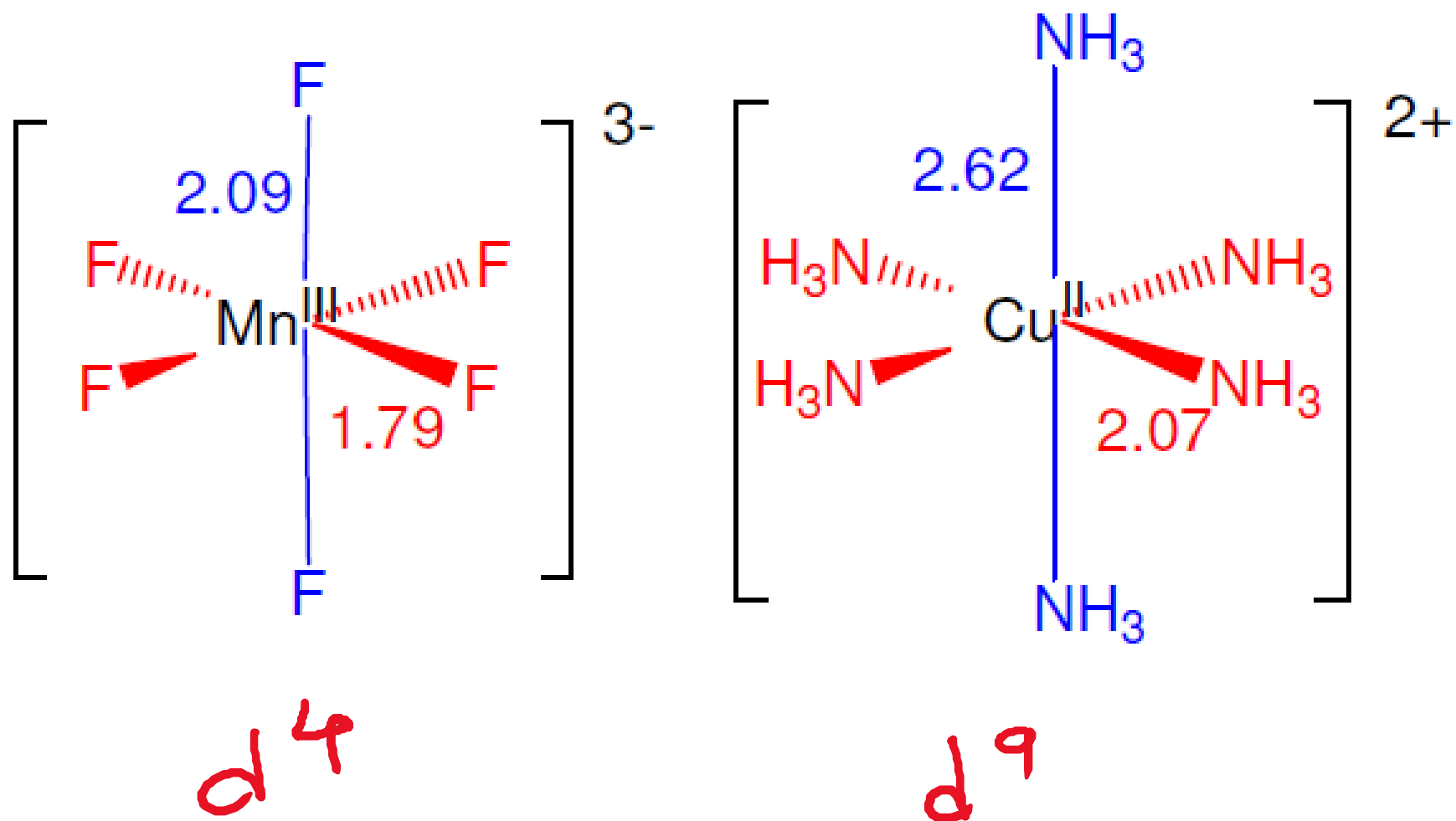
if $P > \Delta_0 \longrightarrow t_{2g}^3 e_g^1$ High Spin

if $P < \Delta_0 \longrightarrow t_{2g}^4 e_g^0$ low spin

d^n	Weak field			Strong field		
	Configuration	Unpaired electrons	CFSE	Configuration	Unpaired electrons	CFSE
d^1	t_{2g}^1	1	$0.4\Delta_o$	t_{2g}^1	1	$0.4\Delta_o$
d^2	t_{2g}^2	2	$0.8\Delta_o$	t_{2g}^2	2	$0.8\Delta_o$
d^3	t_{2g}^3	3	$1.2\Delta_o$	t_{2g}^3	3	$1.2\Delta_o$
d^4	$t_{2g}^3 e_g^1$	4	$0.6\Delta_o$	t_{2g}^4	2	$1.6\Delta_o$
d^5	$t_{2g}^3 e_g^2$	5	$0.0\Delta_o$	t_{2g}^5	1	$2.0\Delta_o$
d^6	$t_{2g}^4 e_g^2$	4	$0.4\Delta_o$	t_{2g}^6	0	$2.4\Delta_o$
d^7	$t_{2g}^5 e_g^2$	3	$0.8\Delta_o$	$t_{2g}^6 e_g^1$	1	$1.8\Delta_o$
d^8	$t_{2g}^6 e_g^2$	2	$1.2\Delta_o$	$t_{2g}^6 e_g^2$	2	$1.2\Delta_o$
d^9	$t_{2g}^6 e_g^3$	1	$0.6\Delta_o$	$t_{2g}^6 e_g^3$	1	$0.6\Delta_o$
d^{10}	$t_{2g}^6 e_g^4$	0	$0.0\Delta_o$	$t_{2g}^6 e_g^4$	0	$0.0\Delta_o$

^a This table is somewhat simplified because pairing energies and electron-electron effects have been neglected.

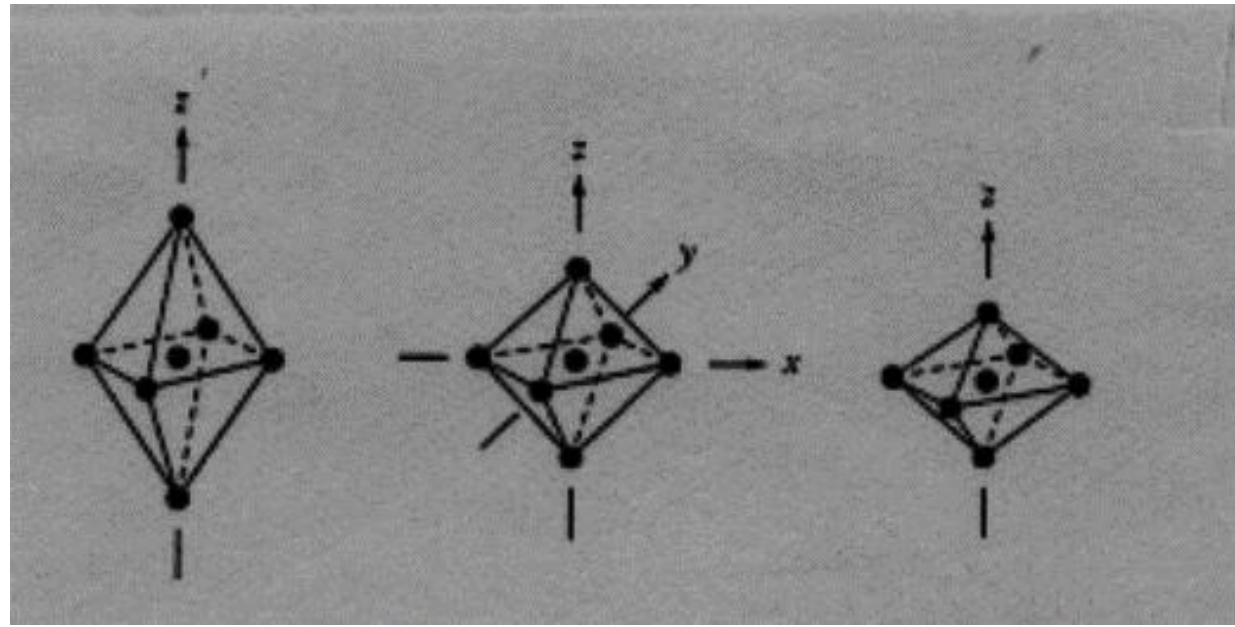




The Jahn-Teller Theorem:

For a non-linear molecule in an electronically degenerate state, distortion must occur to lower the symmetry, remove the degeneracy and lower the energy.

The Jahn-Teller effect is more pronounced when the e_g level is not completely filled



dx^2-y^2

dz^2

dx^2

dxz, dyz

—

—

—

==

e_g

=

t_{2g}

==

—

—

==

—

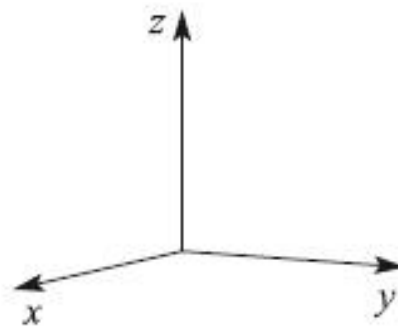
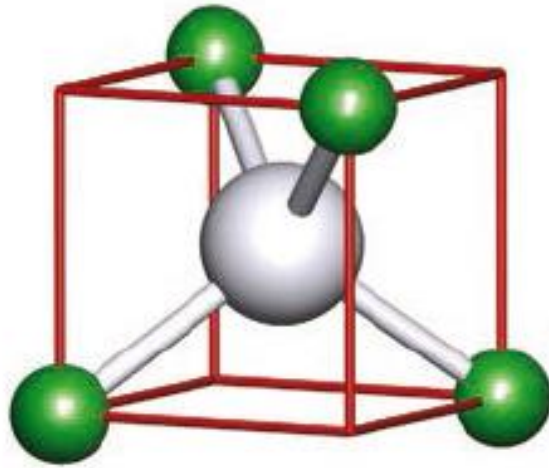
dz^2

dx^2-y^2

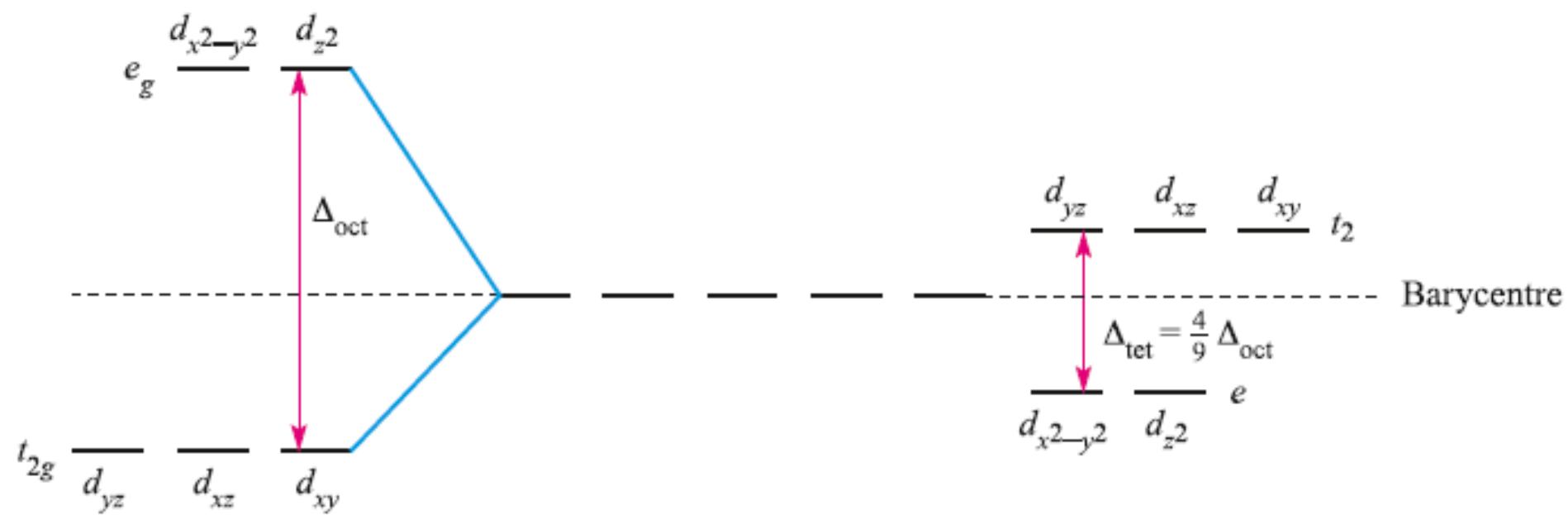
dxz, dyz

dx^2

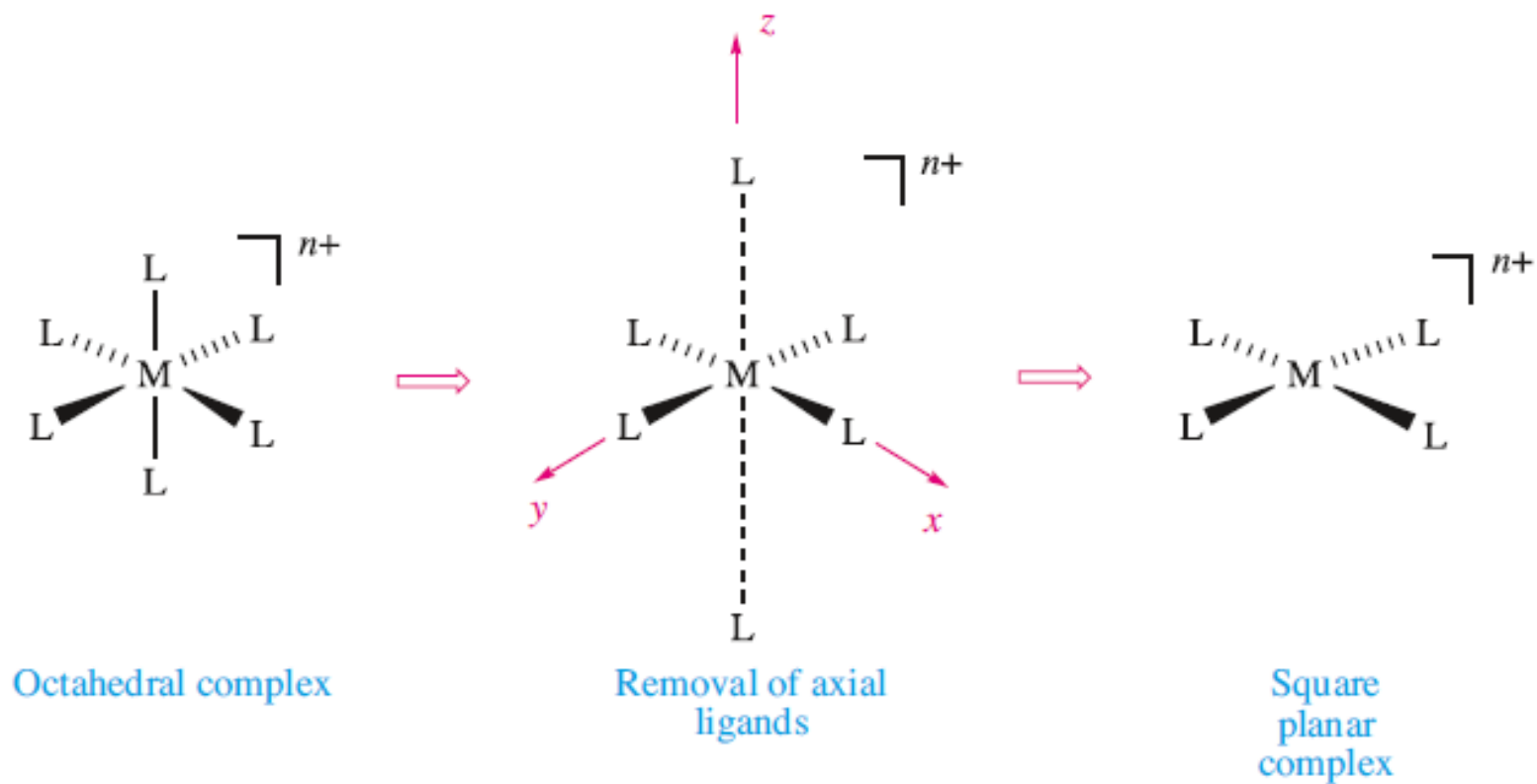
The tetrahedral crystal field

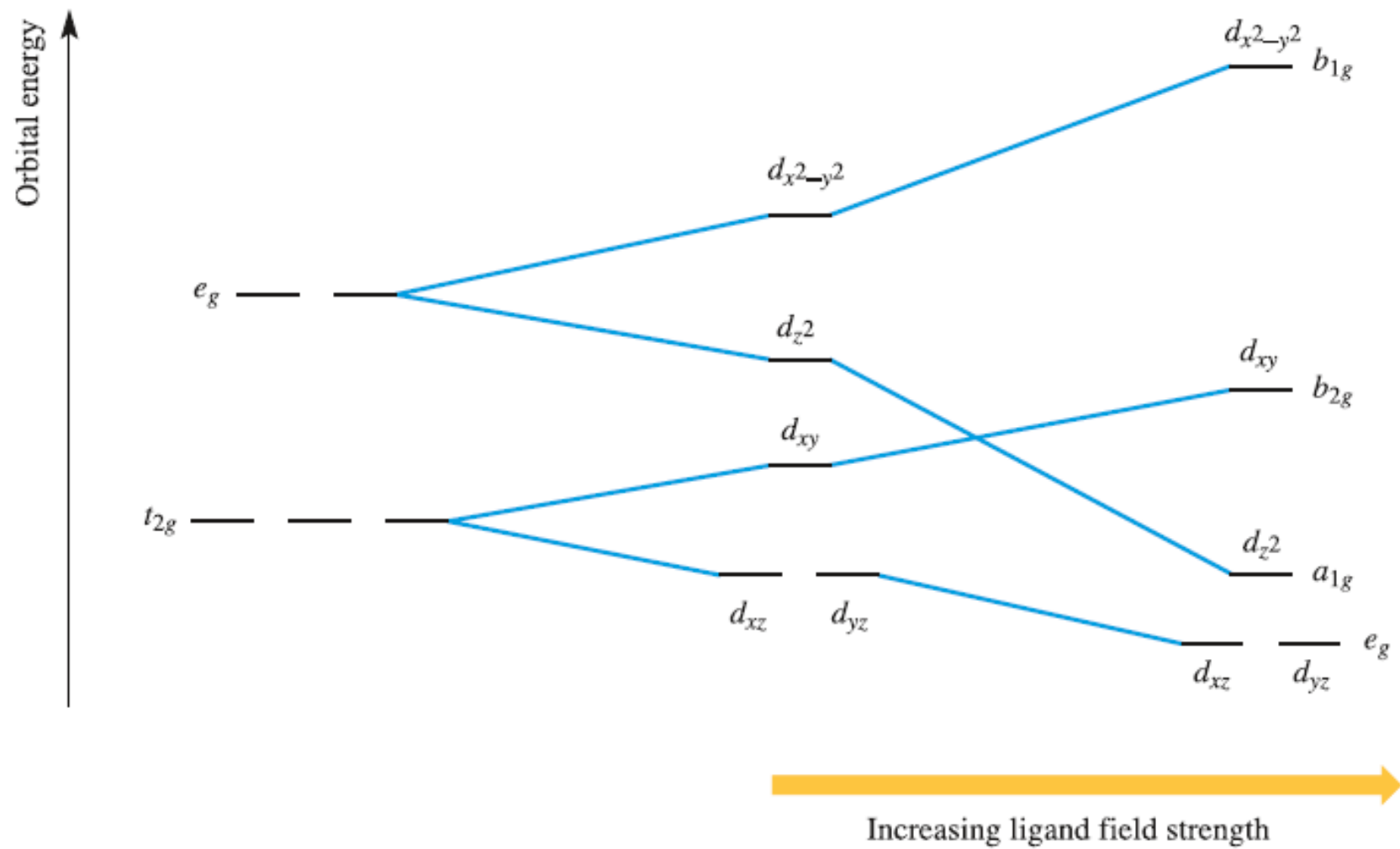


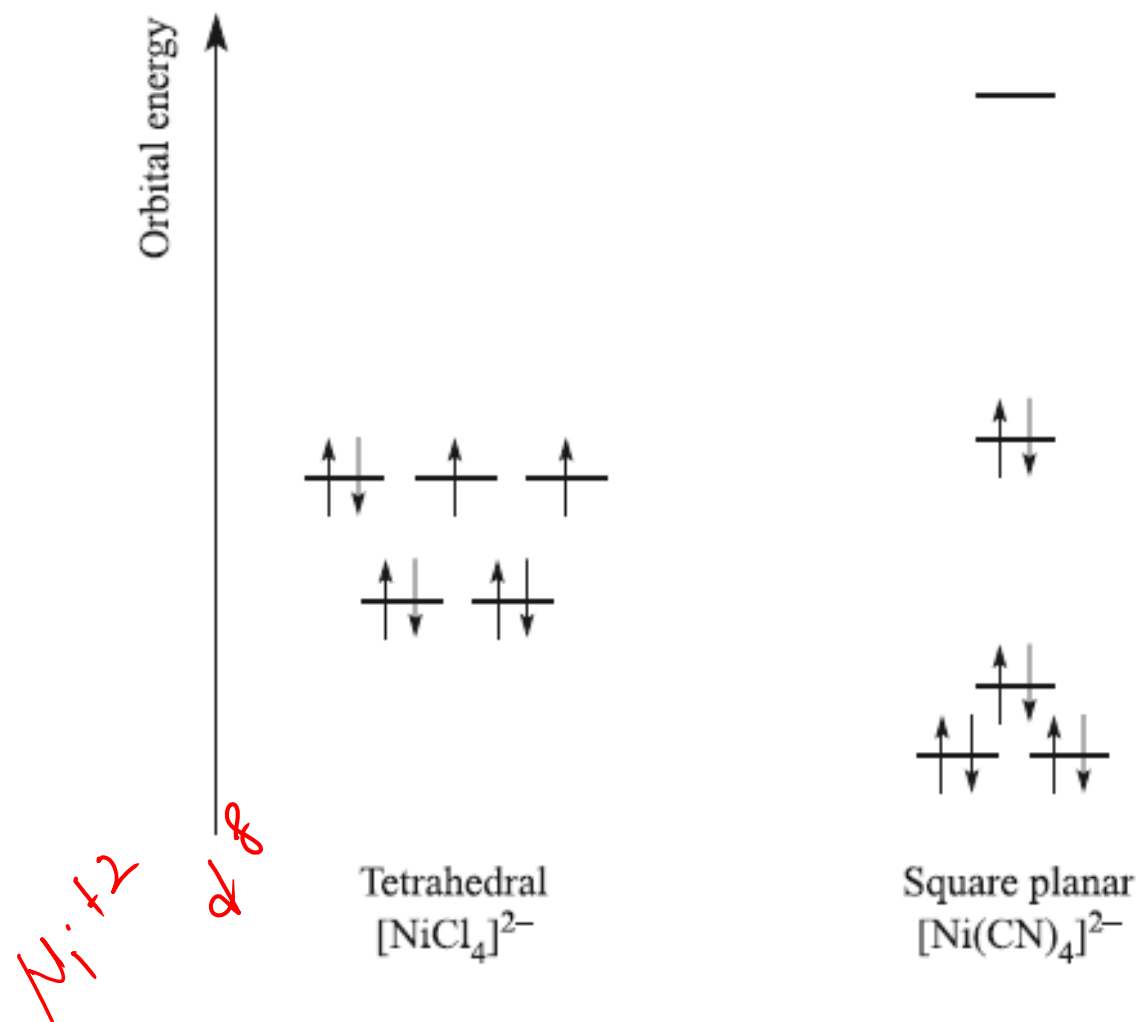
Orbital
energy ↑

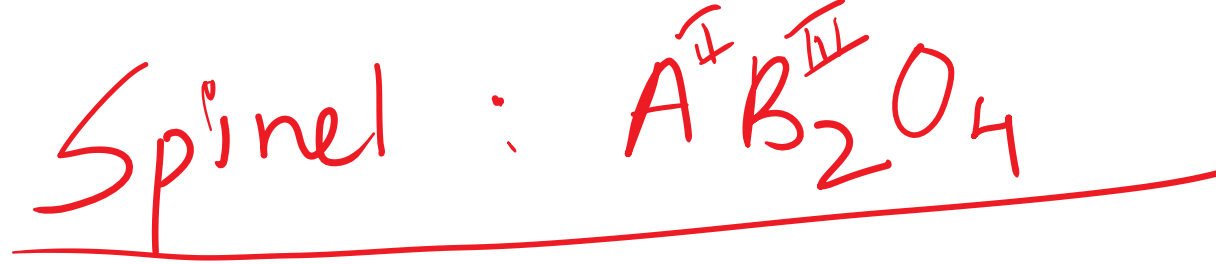


The square planar crystal field







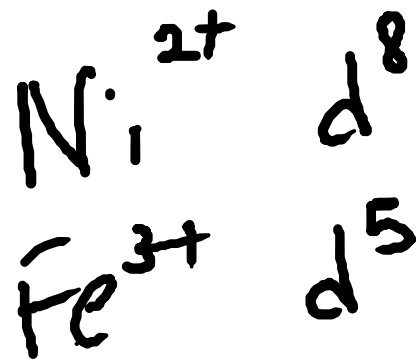


The distribution of metal ions between tetrahedral and octahedral sites in a spinel can be rationalized in terms of CFSEs.

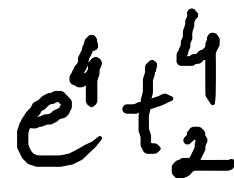
In a normal spinel the tetrahedral sites are occupied by the A^{2+} ions and the octahedral sites by B^{3+} ions: $(A^{\text{II}})^{\text{tet}} (B^{\text{III}})^{\text{oct}} O_4$

In an inverse spinel, the distribution is





Ni²⁺ In tetrahedral



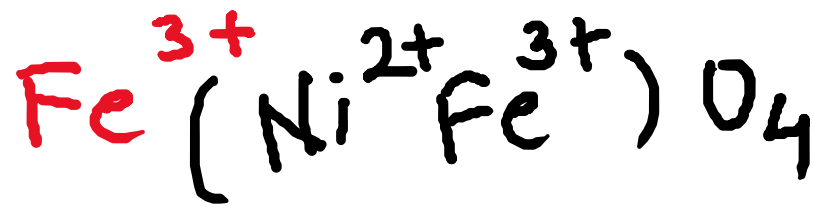
CFSE 0.8 Δ_t

Ni²⁺ In octahedral



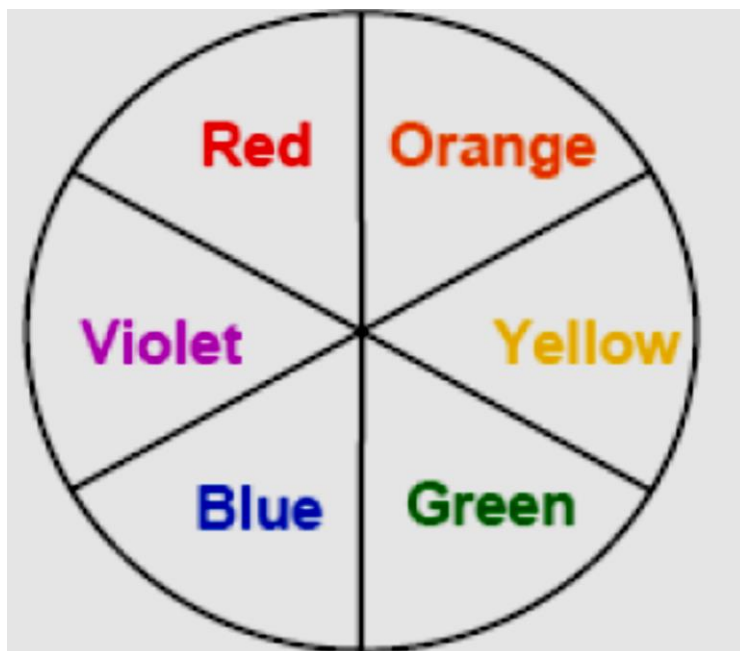
~ 0.4 Δ_o

t_{2g}⁶ e_g² CFSE 1.2 Δ_o



Color



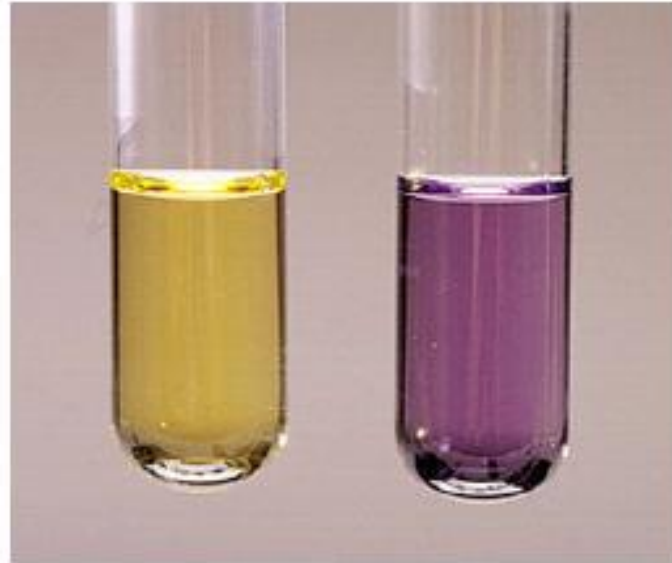
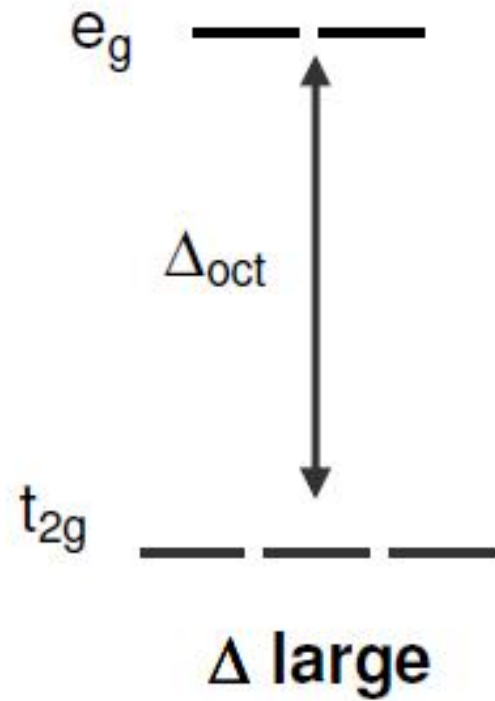


Colour of light absorbed	Approx. λ ranges / nm	Colour of light transmitted
Red	700-620	Green
Orange	620-580	Blue
Yellow	580-560	Violet
Green	560-490	Red
Blue	490-430	Orange
Violet	430-380	Yellow

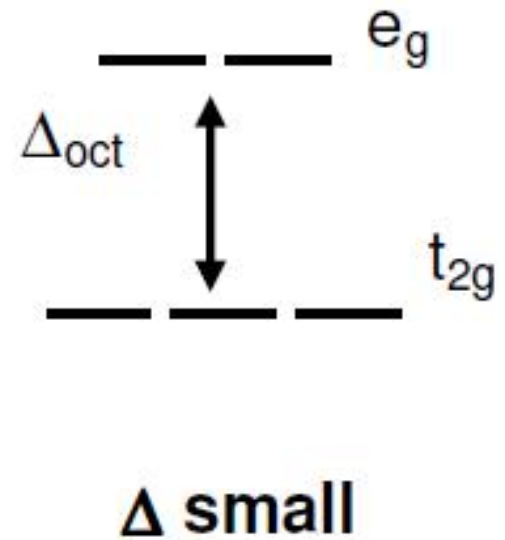
Color of Coordination Complexes

The color of coordination complexes arises from electronic transitions between levels whose spacing corresponds to the wavelengths available in the visible light. In complexes, these transitions are frequently referred to as d-d transitions because they involve the orbitals that are mainly d in character (for examples: t_{2g} and e_g for the octahedral complexes and e and t_2 for the tetrahedral complexes).

violet light absorbed
complex appears yellow



yellow light absorbed
complex appears violet

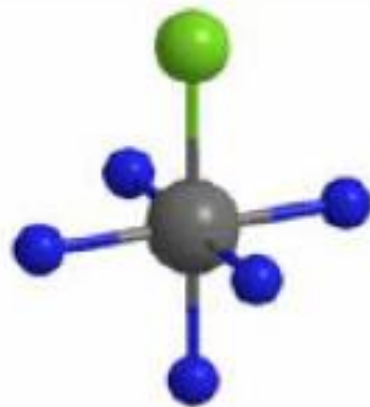


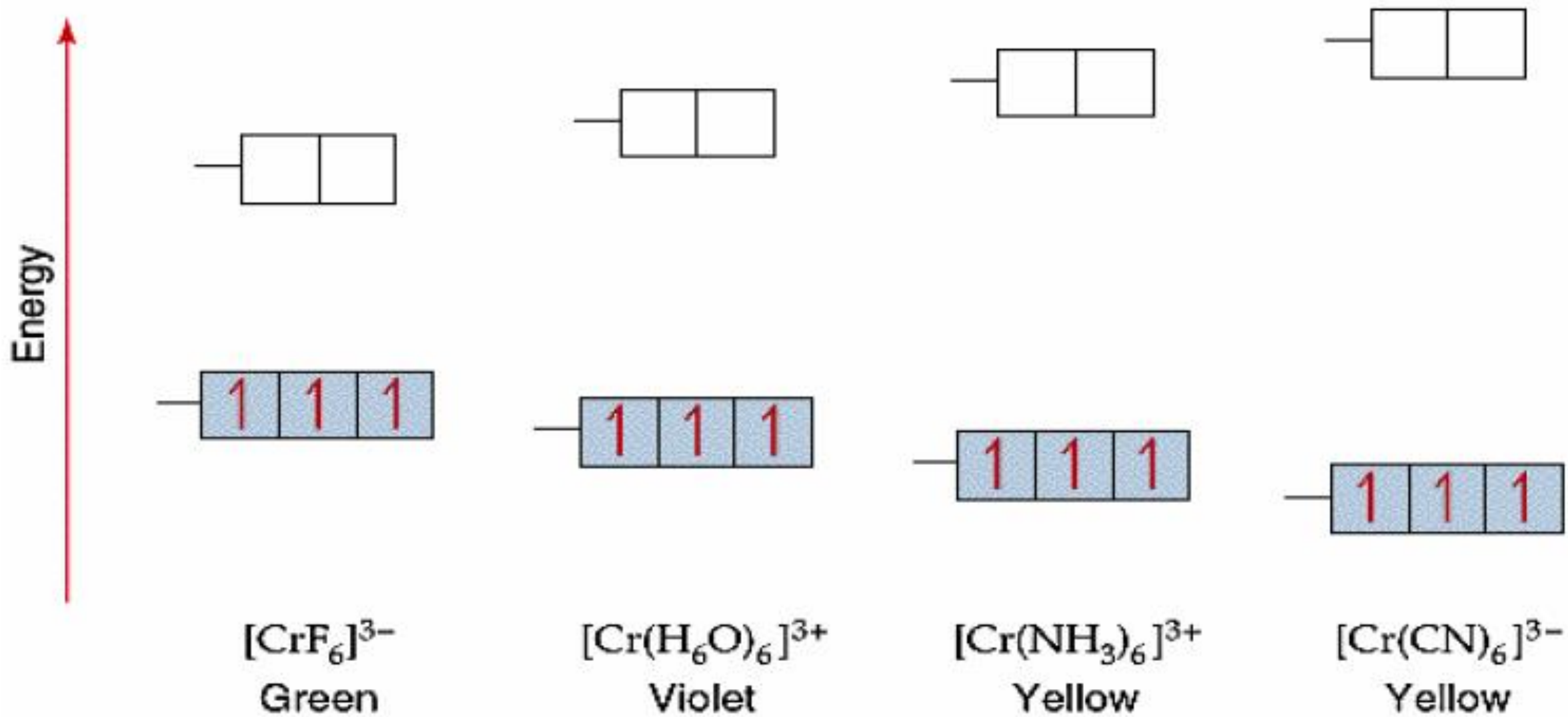


Strong Δ_{oct}
High energy,
violet light absorbed
Complex: yellow

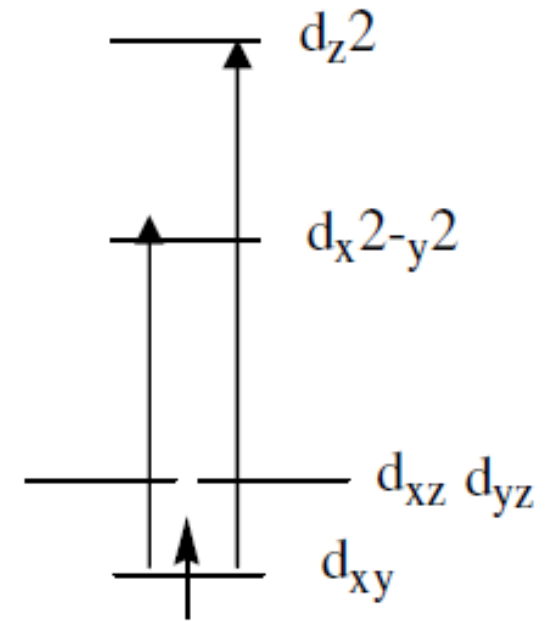
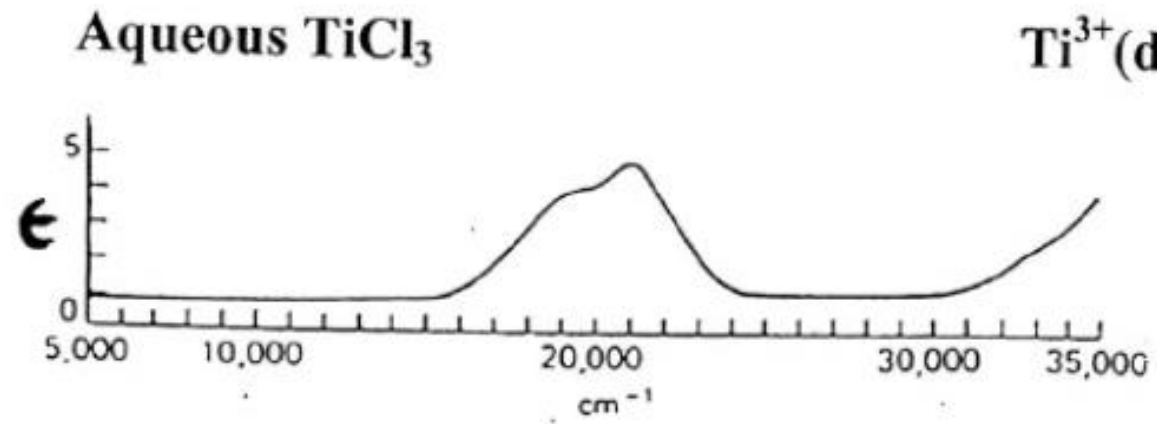


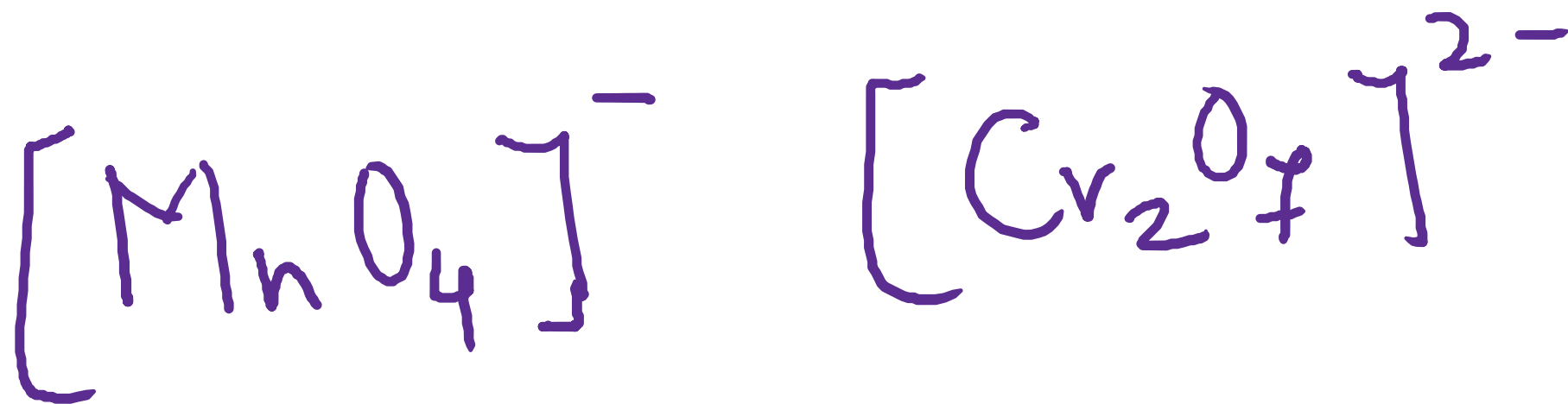
Weak Δ_{oct}
Low energy,
yellow light absorbed
Complex: magenta





appearance of a shoulder in the case of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$





Selection rules for electronic transitions

The Beer-Lambert Law

$$A = \log_{10}(I_0/I) = \epsilon c l$$

where ϵ is the molar extinction coefficient (in $\text{L cm}^{-1} \text{ mole}^{-1}$),

c is concentration in mole L^{-1}

l is the path length in cm.

A is known as 'Absorbance'

The Laporte Rule.

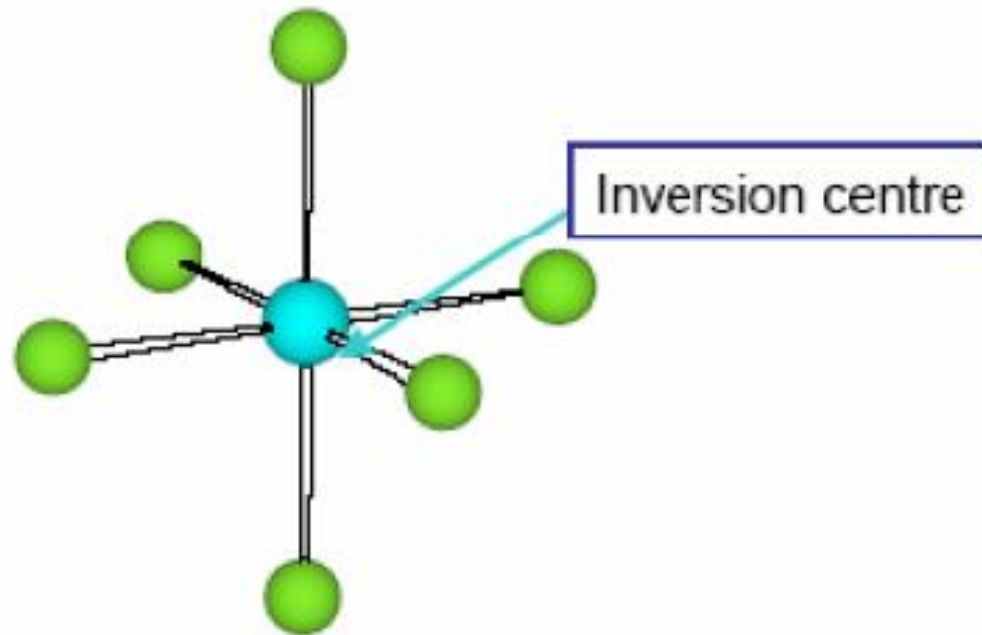
In a molecule or ion possessing a centre of symmetry, transitions are not allowed between orbitals of the same parity, for example d to d. In other words, there must be change in parity ($\Delta l = \pm 1$), i.e. the orbital quantum number should differ by 1.

The rule is not applicable to tetrahedral complexes as it does not contain a center of symmetry.

Octahedral complexes

(Centrosymmetric)

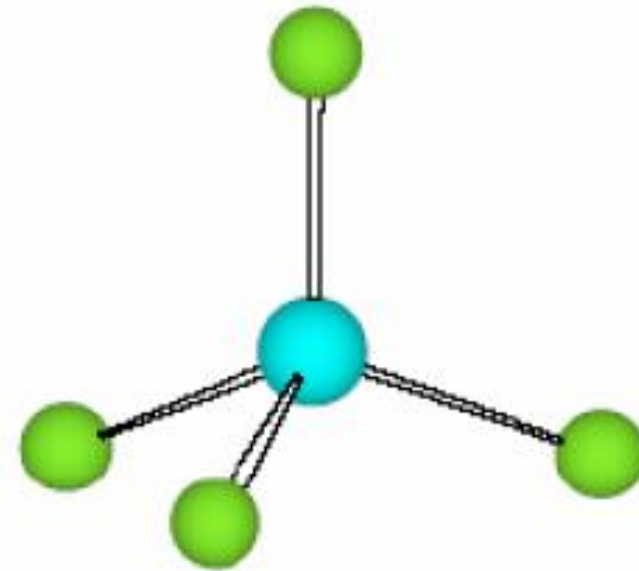
Laporte rule applies



Tetrahedral complexes

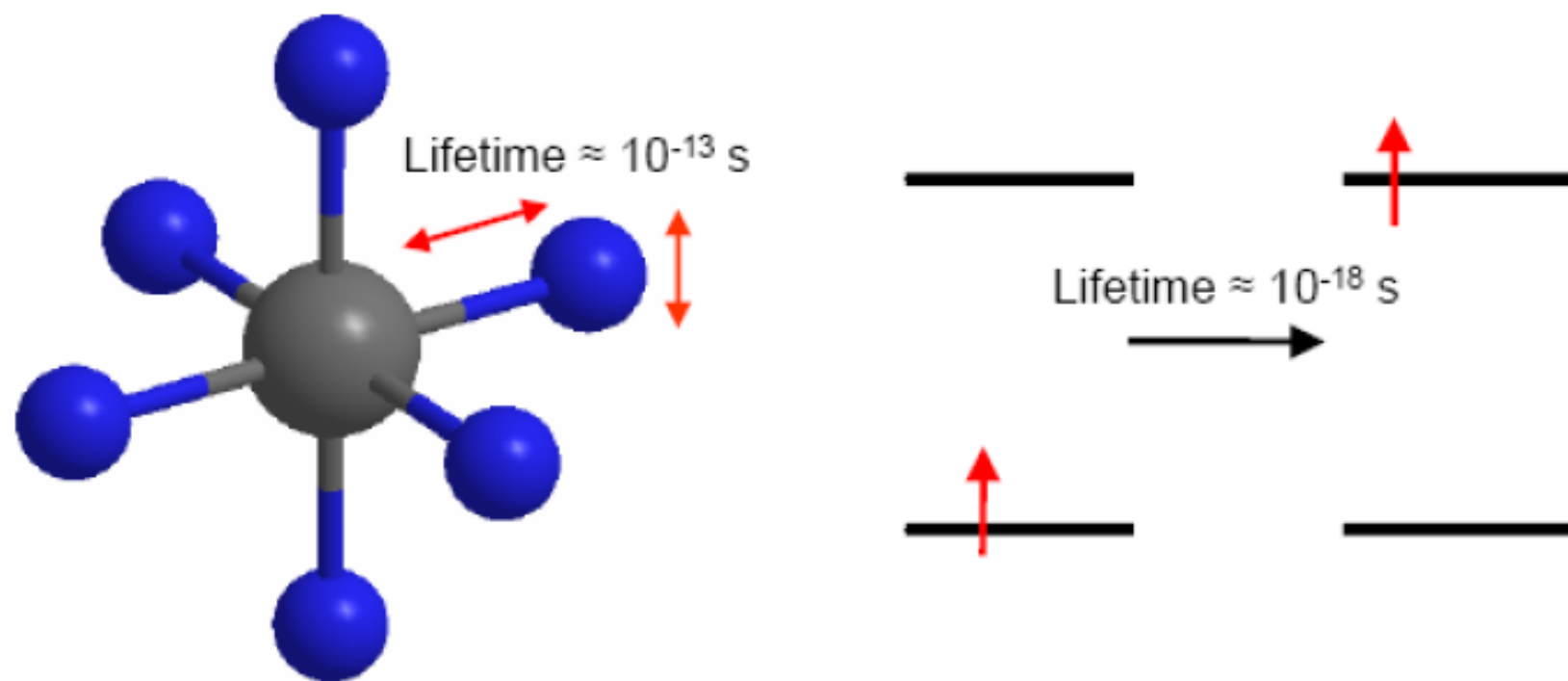
(Non-centrosymmetric)

Laporte rule relaxed



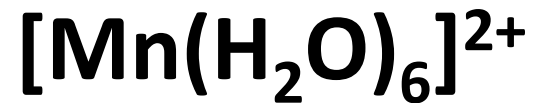
$\epsilon (T_d)$ is a 100 times greater than $\epsilon (O_h)$

Molecular vibrations: Vibronic coupling

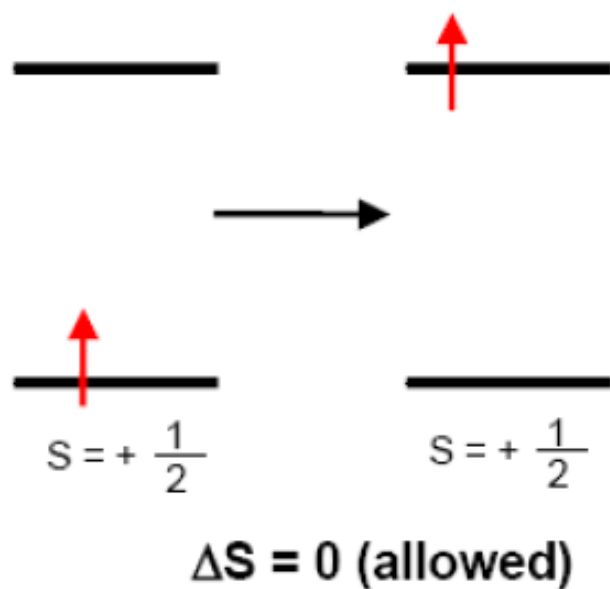


Spin Allowed - Spin Forbidden

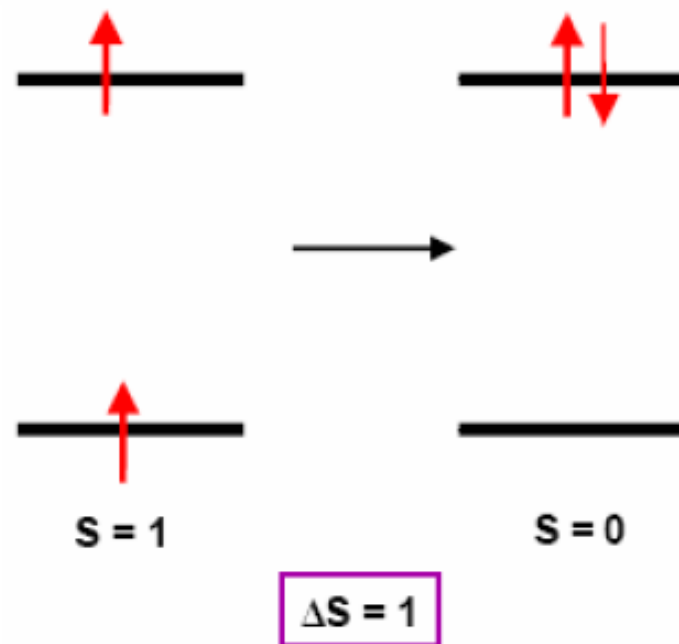
Any transition for which $\Delta S \neq 0$ is *strongly* forbidden; that is, in order to be allowed, a transition must involve no change in spin state.



Promotion of an electron can only proceed if the spin orientation is conserved, i.e. $\Delta S = 0$



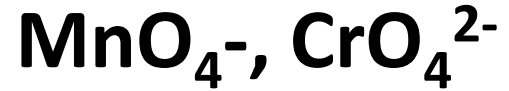
'forbidden' transitions



Charge-Transfer (CT) Bands.

Similar to d-d transitions, charge-transfer (CT) transitions also involve the metal d-orbitals. CT bands are observed if the energies of empty and filled ligand- and metal-centered orbitals are similar. The direction of the electron transfer is determined by the relative energy levels of these orbitals:

Ligand to-Metal charge transfer (LMCT)



For MnO_4^- , the d-electron count on Mn(VII) is d^0

The origin of the color in this species is not due to d-d transition, rather, charge transfer from O_2^- to Mn(VII), described as LMCT band.

Metal-to Ligand charge transfer (MLCT)



In this complex the charge transfer occurs from Fe(II) to the empty π^* orbitals of bpy ligand

LMCT $\Rightarrow$ favourable when metal centre is in high oxidation state

MLCT $\Rightarrow$ favoured when electron rich metal centre is present.

Expected intensities of electronic transitions

<u>Type of transition</u>	<u>ϵ value $\text{dm}^3 \text{mol}^{-1} \text{cm}^{-1}$</u>	<u>Example</u>
spin forbidden d-d	< 1	$[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ High spin d^5

Laporte forbidden
Spin Allowed
d-d

1 - 10 centrosymmetric
 $[\text{Ti}(\text{H}_2\text{O})_6]^{+3}$

10 - 1000 Non centrosymmetric

charge transfer
fully Allowed

1000 - 50000 MnO_4^-